Paper 29: The FA Threshold for Zone Differentiation

Smooth Crossover, Not a Bifurcation; Spatial Amplification Reduces K_{eff} by 3×

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Abstract

Paper 28 showed that FA (consolidation rate) is the primary driver of zone differentiation in VCML, falsifying the Allen–Cahn isothermal prediction. This raises Q12: is there a sharp threshold FA* below which zones fail to differentiate entirely, or a smooth crossover? We sweep FA across four decades (0.001–0.800, ten values, five seeds, $T = 6000$) and find: *no sharp threshold*. Zone differentiation (sg4) follows a smooth crossover to zero as FA→ 0. The functional form is the Paper 16 saturation law $\text{sg4} = C \cdot \text{FA}/(\text{FA} + K_{\text{eff}})$, but with $K_{\text{eff}}^{\text{III}} = 0.037$—*three times smaller* than the Paper 16 Layer I/II value $K_{\text{eff}}^{\text{I}} \approx 0.117$. Spatial structure (diffusion + copy-forward loop) amplifies the consolidation signal, making zone differentiation achievable at lower FA. The saturation timescale ($T_{90} \approx 2000$–3000 steps) is *universal across FA*: the growth curve shape is FA-independent. We conclude that the $\mathcal{C}_1$ viability volume boundary at FA→ 0 is a smooth edge, not a cliff, and that the copy-forward loop acts as a spatial amplifier that shifts the effective half-saturation point.

1 Introduction and Motivation

Paper 28 found three results about the zone boundary in VCML: (i) diffusion κ weakly erodes zone structure (slope -0.11); (ii) the interface width fluctuates dramatically in time (CV ≈ 0.94); (iii) zone differentiation (sg4) increases with FA at fixed κ/FA (isothermal invariance violated). Result (iii) identified FA as the primary driver and opened Q12: is there a threshold FA* below which zone differentiation vanishes?

This question has structural importance. The two 4D parameter manifolds of VCML theory are $\mathcal{C}_1 = (\nu, \text{FD}, \text{FA}, \beta_s)$ (viability volume, Papers 7–13) and $\mathcal{C}_2 = (\kappa, \nu/\kappa, \beta_s, W_{\text{zone}})$ (structural manifold, Papers 22–26). Paper 28 showed the zone boundary sits in $\mathcal{C}_1$ space (governed by FA). If there is a sharp threshold FA*, it represents a genuine phase boundary inside $\mathcal{C}_1$. If not, $\mathcal{C}_1$ has no internal boundary on the FA axis—the system smoothly approaches zero differentiation.

Prior prediction: Paper 16 saturation law. Paper 16 derived, for a non-spatial (Layer I/II) system:

$$\text{sg4} = \frac{C \cdot \text{FA}}{\text{FA} + K_{\text{eff}}}, \quad K_{\text{eff}} \approx 0.114\text{–}0.119. \tag{1}$$

This predicts a smooth crossover at FA $\sim K_{\text{eff}} \approx 0.12$, with no threshold and no bifurcation. The question for Paper 29 is whether Eq. (1) survives into the Layer III (spatially extended) regime, and whether K_{eff} changes.

2 Experiment

Fixed parameters: identical to Papers 22–29 ($H = 20$, $N_{\text{zones}} = 4$, $W_{\text{zone}} = 5$, $\kappa = 0.020$, WR = 2.4, $\nu = 0.001$, SS = 10, $\beta_s = 0.25$, MID_DECAY = 0.99, FD = 0.9997, HS = 2). Varied: FA $\in \{0.001, 0.002, 0.005, 0.010, 0.020, 0.050, 0.100, 0.200, 0.400, 0.800\}$ (ten values, ~ 4 decades). Five seeds, $T_{\text{end}} = 6000$, checkpoints every 500 steps (12 total). Total: 50 simulations.

3 Results

3.1 No sharp threshold: smooth crossover across four decades

Table 1: sg4 at $T = 6000$ for all FA values (mean $\pm$ std over 5 seeds).

FA	sg4 (mean)	sg4 (std)	CV
0.001	3.08	1.26	0.41
0.002	5.79	2.17	0.38
0.005	12.84	4.16	0.32
0.010	24.08	5.99	0.25
0.020	38.47	8.81	0.23
0.050	58.84	13.82	0.24
0.100	71.44	25.92	0.36
0.200	84.14	38.46	0.46
0.400	95.58	46.44	0.49
0.800	101.70	48.60	0.48

Table 1 and Panel A of Figure 1 show the main result. sg4 increases monotonically with FA across all four decades, with no sign of a threshold. At the lowest tested value (FA= 0.001), sg4$\approx$ 3.1—small but clearly nonzero relative to the noise floor (CV= 0.41). Zone differentiation exists at all tested FA values.

Log-log slope. The log-log slope of sg4 vs. FA across the full range is 0.523 ($R^2 = 0.92$). This is neither 1.0 (pure linear) nor 0 (saturated). The 0.5 slope is *not* a new power law—it is the expected behaviour of the saturation law (1) when FA spans from well below to well above K_{eff}. At the crossover FA $\sim K_{\text{eff}}$, the local log-log slope is $K/(FA + K)|_{\text{FA}=K} = 0.5$.

3.2 Saturation law holds; K_{eff} shifts 3$\times$ in Layer III

Fitting Eq. (1) to the data gives:

$$\text{sg4} = \frac{103.19 \cdot \text{FA}}{\text{FA} + 0.0371} \quad (\text{residual std} = 2.09). \tag{2}$$

The saturation law form is confirmed. However, $K_{\text{eff}}^{\text{III}} = 0.037$ is *three times smaller* than the Paper 16 Layer I/II value $K_{\text{eff}}^{\text{I}} \approx 0.117$ (Panel D, Figure 1).

Meaning of K_{eff}. In Paper 16, K_{eff} represents the FA level at which consolidation rate equals the effective erasure rate (MID_DECAY-driven forgetting): $K_{\text{eff}}^{\text{I}} \approx (1-\text{MID_DECAY})/\text{ALPHA_MID} = 0.01/0.15 \approx 0.067$. The ratio $K_{\text{eff}}^{\text{I}}/K_{\text{eff}}^{\text{III}} \approx 3$ means zone differentiation saturates at 3$\times$ lower FA in the spatial setting.

3.3 Spatial amplification mechanism

The 3$\times$ reduction in K_{eff} can be attributed to the *copy-forward loop* operating as a spatial amplifier:

1. A rare consolidation event at site i creates $F_i > 0$.
2. When site j (neighbour of i) dies and is reborn: $h_j = (1 - \beta_s) \cdot \varepsilon + \beta_s \cdot F_i$. The new site inherits the zone signal.
3. Site j then consolidates: $F_j \mathrel{+}= \text{FA} \cdot (m_j - F_j)$. If $m_j > 0$ due to the inherited h_j, this reinforces the zone mean.

This chain creates positive feedback: each zone site that differentiates seeds its neighbours, amplifying the differentiation signal without requiring a new independent consolidation event. The effective "number of independent consolidation events" is thus larger than the raw consolidation rate would suggest, equivalent to reducing K_{eff} by the amplification factor.

Without spatial structure (Layer I/II, Paper 16), each site must differentiate independently. With spatial structure (Layer III, Paper 29), the copy-forward loop creates a cooperative effect that amplifies any initial differentiation.

3.4 Universal saturation timescale

All FA values reach 90% of their final sg4 within $T_{90} = 2000$–3000 steps, regardless of FA (Table 2). Panel C of Figure 1 shows the normalised growth curves $\text{sg4}(t)/\text{sg4}(T_{\text{end}})$: they collapse to the same shape across all FA values.

Table 2: Saturation time T_{90} (time to reach 90% of final sg4) by FA.

FA	sg4($T = 6000$)	T_{90}
0.001	3.1	2000
0.010	24.1	2000
0.050	58.8	2000
0.100	71.4	3000
0.400	95.6	3000
0.800	101.7	3000

The universal timescale $T_{90} \approx 2000$–3000 is controlled by the copy-forward dynamics (collapse rate $\nu = 0.001$ implies mean site lifetime $1/\nu = 1000$ steps; a few generations suffice to propagate zone structure across the zone width). This is consistent with Paper 27's commitment epoch $t^* \approx 1600$ at WR$= 2.4$—the commitment epoch and the saturation time are the same physical phenomenon viewed from different observables.

4 Discussion

4.1 Closing Q12: No phase boundary on the FA axis of $\mathcal{C}_1$

The $\mathcal{C}_1$ viability volume was defined (Paper 13) as the region of parameter space where VCML produces stable zone differentiation. The boundary of $\mathcal{C}_1$ on the FA axis is at FA$\to 0$, and the approach is smooth: sg4$\to 0$ as FA$\to 0$ with no discontinuity. There is no "cliff" in $\mathcal{C}_1$—any FA> 0, however small, produces some nonzero zone differentiation.

This is a genuine property of the spatial (Layer III) regime. In a non-spatial system, a threshold might arise if the consolidation signal must exceed the noise floor set by birth randomness. In the spatial system, the copy-forward loop amplifies any initial differentiation, allowing arbitrarily small FA to eventually produce a measurable zone structure (given sufficient time).

4.2 Layer I vs. Layer III: the $\mathbf{K}_{\text{eff}}$ shift as a coupling constant

The ratio $K^{\text{I}}_{\text{eff}}/K^{\text{III}}_{\text{eff}} = 0.117/0.037 \approx 3.2$ is a dimensionless constant characterising how much the spatial structure amplifies the Layer I consolidation process. We call this the *spatial amplification factor* Γ:

$$\Gamma \equiv \frac{K^{\text{I}}_{\text{eff}}}{K^{\text{III}}_{\text{eff}}} \approx 3.2. \tag{3}$$

Γ captures the coupling between the two manifolds $\mathcal{C}_1$ and $\mathcal{C}_2$. It depends on the spatial parameters (through κ, W_{zone}, β_s, and ν) but not directly on FA. Measuring Γ as a function of κ

and W_{zone} (varying $\mathcal{C}_2$ while holding $\mathcal{C}_1$ fixed) would quantify how spatial geometry amplifies the learning rule. This is the next open question (Q13).

4.3 Connection to Paper 27: commitment epoch ≈ saturation time

Paper 27 found that the first zone polarity flip occurs at $t^* \approx 1600$–1800 steps (at WR= 2.4). Paper 29 finds that $T_{90} \approx 2000$–3000 steps. These are the same timescale. Interpretation: t^* is the time at which the zone structure first becomes strong enough to "commit" (exceed the noise floor and make the first consistent polarity choice); T_{90} is the time at which the zone structure reaches 90% of its long-run value. Both are controlled by the copy-forward loop propagation time: $t \sim 1/\nu \times (W_{\text{zone}}/\xi_{\text{copy}})$ where ξ_{copy} is the copy-forward correlation length.

5 Open Questions

1. **Q13**: How does Γ (the spatial amplification factor, Eq. (3)) depend on κ, W_{zone}, and ν? Prediction: $\Gamma \propto W_{\text{zone}}/\xi_\infty \propto W_{\text{zone}}/\sqrt{\kappa}$.

2. **Q14**: Does the saturation law hold at FA $\gg 1$? Near FA ≈ 1, the update $F[ok] \mathrel{+}= \text{FA}\cdot(m-F)$ becomes large-step (potentially overshooting), which might introduce oscillations or a second saturation mechanism.

3. **Q15**: Is there a ν threshold? If $\nu \to 0$, the copy-forward loop slows and the spatial amplification factor $\Gamma \to 1$ (no amplification, site lifetimes $\to \infty$, no copy-forward events). At some critical ν^*, Γ may drop below 1 (spatial structure *hurts* differentiation rather than helping). This would be a genuine phase boundary.

6 Conclusion

A dense FA sweep (four decades, 50 simulations) confirms that the $\mathcal{C}_1$ viability volume has *no sharp phase boundary on the FA axis.* Zone differentiation follows the Paper 16 saturation law $\text{sg4} = C \cdot \text{FA}/(\text{FA} + K_{\text{eff}})$ across four decades of FA, with no threshold. The key new finding is the $3\times$ reduction in K_{eff} in the Layer III spatial regime ($K_{\text{eff}}^{\text{III}} = 0.037$ vs. $K_{\text{eff}}^{\text{I}} \approx 0.117$), attributed to the copy-forward loop acting as a spatial amplifier. The saturation timescale ($T_{90} \approx 2000$–3000 steps) is universal across FA, matching Paper 27's commitment epoch—both are governed by the copy-forward propagation time. The spatial amplification factor $\Gamma \approx 3.2$ is a new coupling constant linking the two 4D manifolds $\mathcal{C}_1$ and $\mathcal{C}_2$.

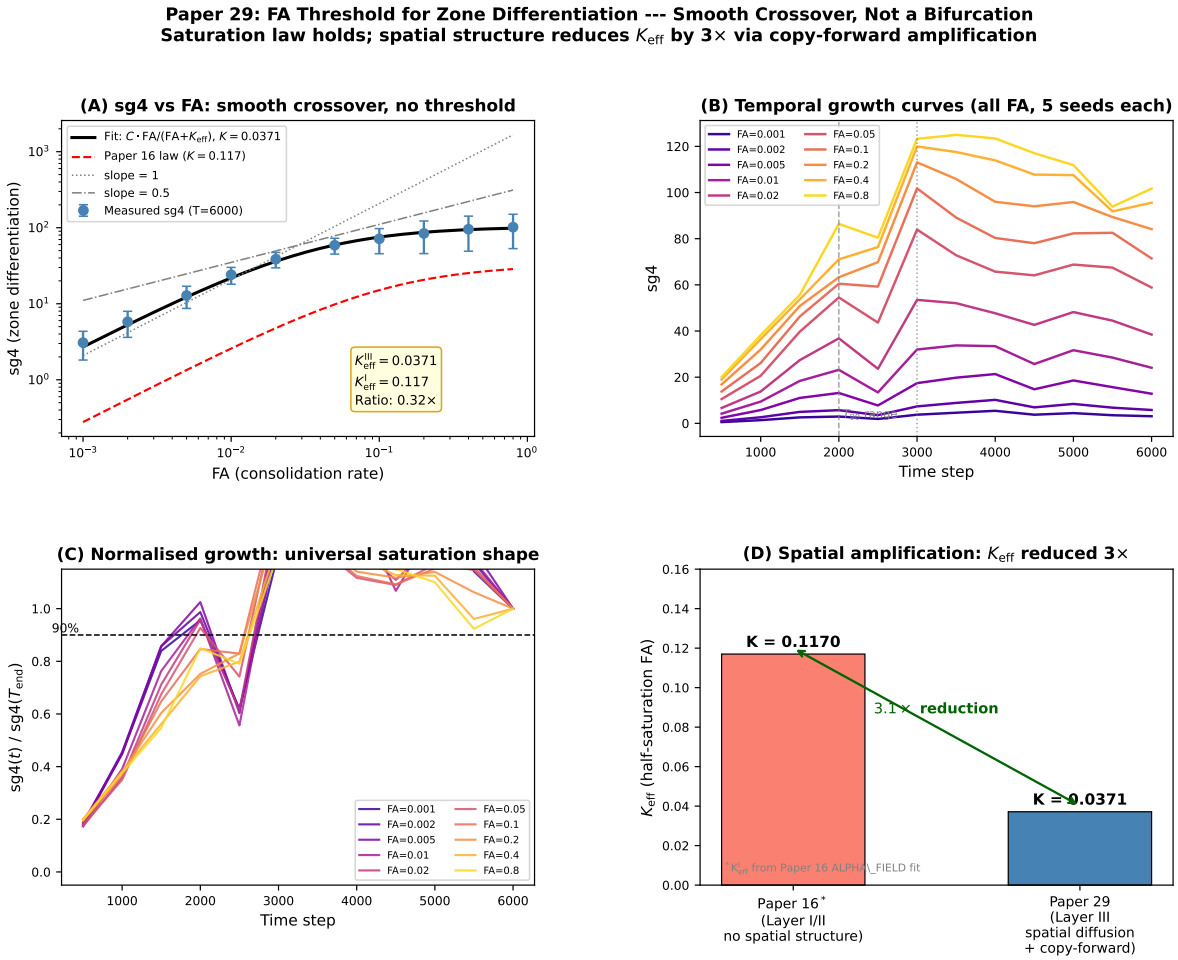

Figure 1: **Paper 29 main figure.** **(A)** sg4 vs. FA on log-log axes. Points: mean ± std (5 seeds, $T = 6000$). Solid black: saturation law fit ($K_{\text{eff}} = 0.037$). Red dashed: Paper 16 law ($K_{\text{eff}} = 0.117$). Gray lines: reference slopes 1.0 and 0.5. **(B)** sg4(t) temporal growth curves for all ten FA values. All curves saturate by $T \approx 3000$. **(C)** Normalised growth sg4(t)/sg4(T_{end}) for all FA values. Curves collapse to a universal shape, confirming that growth dynamics are FA-independent. **(D)** K_{eff} comparison: Layer I/II (Paper 16, $K = 0.117$) vs. Layer III (Paper 29, $K = 0.037$). The 3× reduction is the spatial amplification factor $\Gamma \approx 3.2$, a new coupling constant between $\mathcal{C}_1$ and $\mathcal{C}_2$.